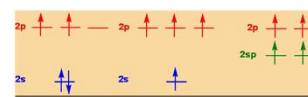
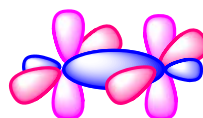
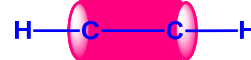
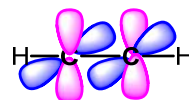
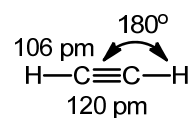


第四章

炔烃和共轭二烯烃

Alkynes and Conjugated Dienes

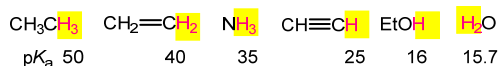
Ethyne



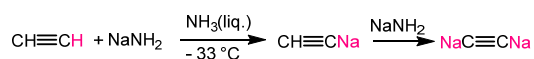
- sp杂化，键角180°，线形分子
- 2个π键，π电子云呈圆柱体

2

Acidity



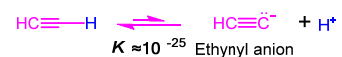
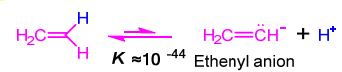
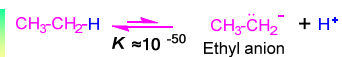
- 末端炔烃酸性比水、醇弱，但比氨强



3

物理性质

Acidity

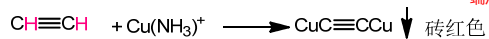
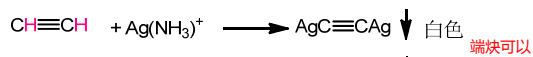
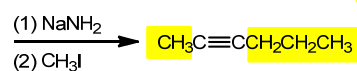
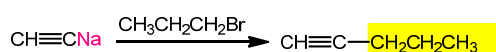


酸性增强

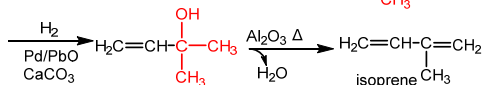
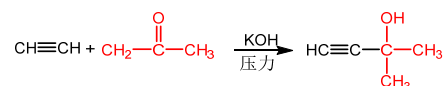
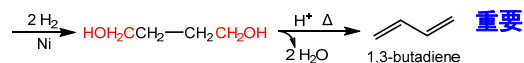
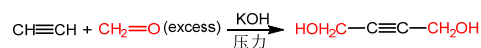
- The **electron-attracting** character of the **sp** carbon leads to the **increased acidity** of the **alkynyl** hydrogen.

- 碳原子的电负性随着s成分增加而增大， $sp^3 < sp^2 < sp$.

4

Acidity: 鉴别及反应**Alkylation of Acetylide Anion**

5

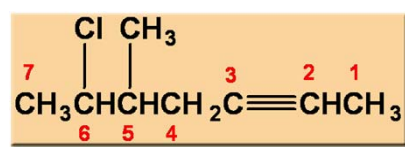
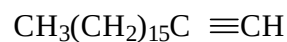
末端炔烃与醛酮的加成

6

炔烃的命名

- 与烯烃的命名原则相同，改“烯”为“炔”
- 分子中同时存在烯键和炔键时，母体名为“烯炔”
- 编号按最低序列规则
- 若编号有选择时以双键位次最小

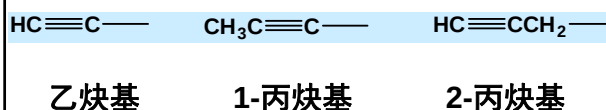
7

Example**6-氯-5-甲基-2-庚炔**

1-十八碳炔

8

炔基



9

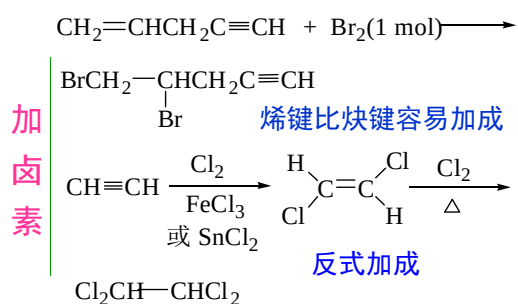
炔烃的化学性质

与烯烃类似，
对比学习

1. 炔烃的酸性
2. 炔烃的亲电加成
3. 炔烃的硼氢化-氧化反应
4. 炔烃的自由基加成
5. 炔烃的加氢和还原
6. 炔烃的亲核加成
7. 炔烃的氧化
8. 乙炔的聚合反应

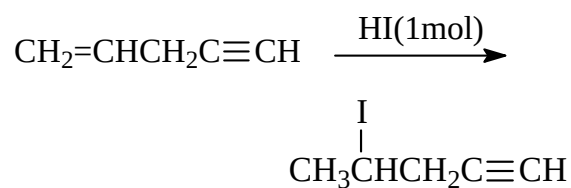
10

炔烃的亲电加成



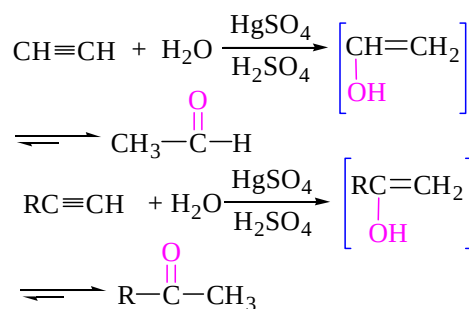
11

Example



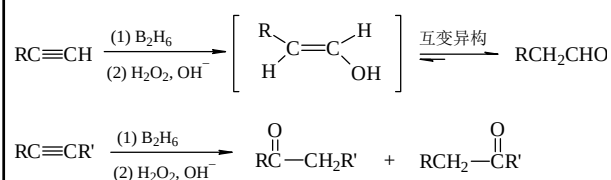
12

加水



13

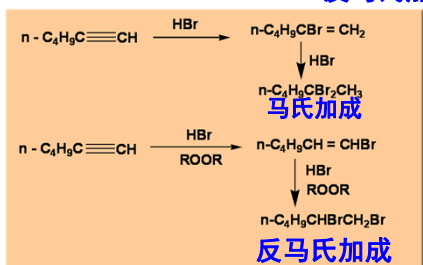
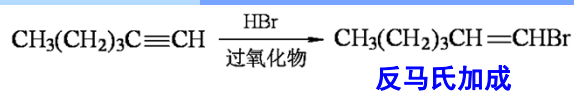
炔烃的硼氢化—氧化反应



末端炔烃经硼氢化—氧化反应生成醛
其它炔烃则生成酮

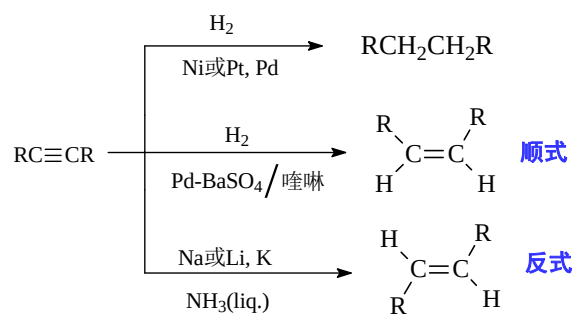
14

炔烃的自由基加成



15

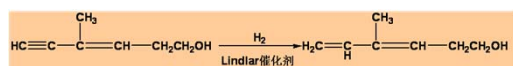
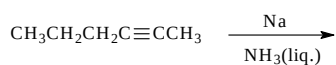
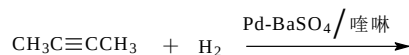
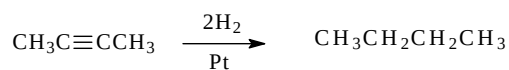
炔烃的加氢和还原



Lindlar 催化剂：
Pd / BaSO₄-喹啉或 Pd / CaCO₃, PbO

16

Example



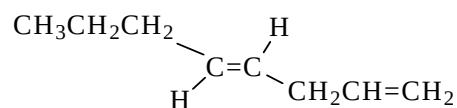
Lindlar 催化剂只对炔烃加氢有效

17

测试：请同学们完成如下题目：

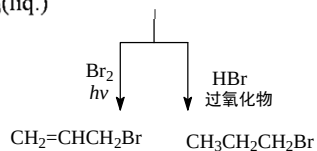
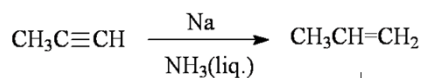
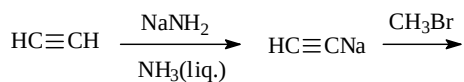
以C2以下的有机化合物（包括C2）为起始原料合成下列化合物

（尝试逆合成分析）



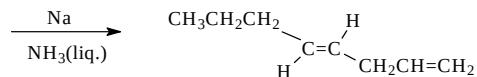
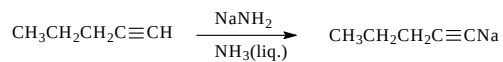
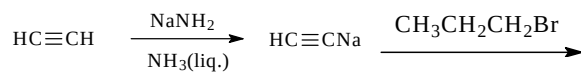
18

合成题目讲解



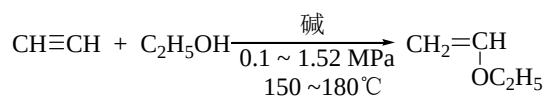
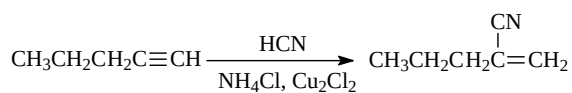
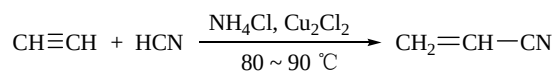
19

合成题目讲解



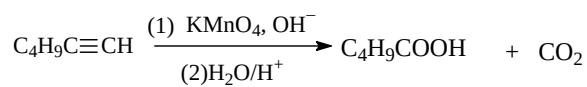
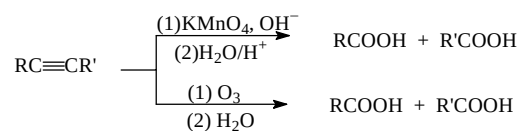
20

炔烃的亲核加成



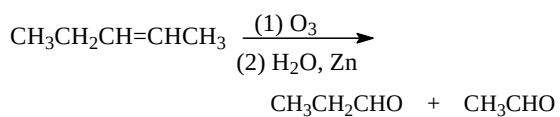
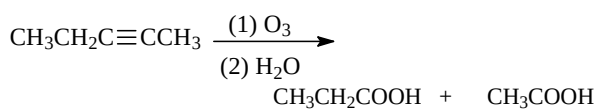
21

炔烃的氧化



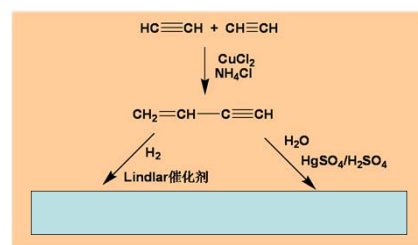
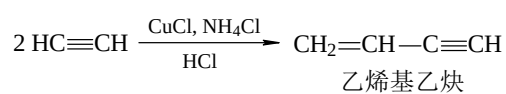
22

Example



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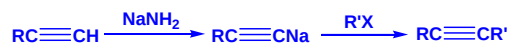
乙炔的聚合



24

炔烃的制备

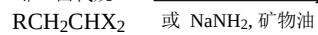
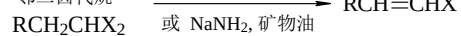
• 炔化物的炔化



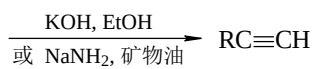
• 二卤代烷去卤化氢



邻二卤代烷

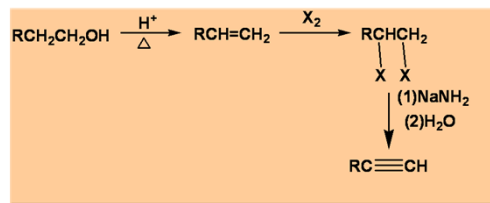


偕二卤代烷



25

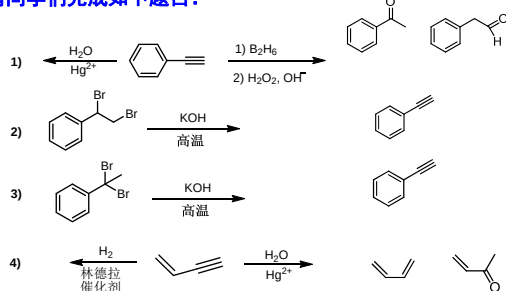
Example



26

测试

请同学们完成如下题目：



共轭二烯

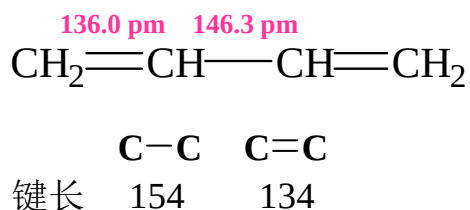
二烯烃的分类

- 累积二烯 $\text{CH}_2=\text{C}=\text{CH}_2$
- 孤立二烯 $\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CH}=\text{CH}_2$
- 共轭二烯 $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$

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共轭二烯的特性

- 结构特性——键长平均化

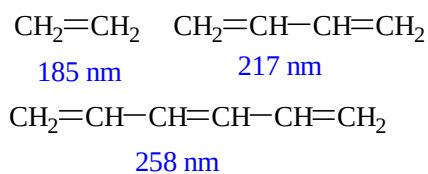


29

共轭二烯的物理特性

- 紫外吸收向长波方向移动；折射率增高；趋于稳定

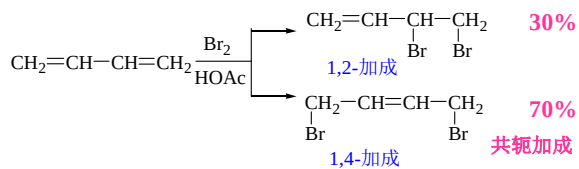
紫外吸收波长



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共轭二烯的化学性质

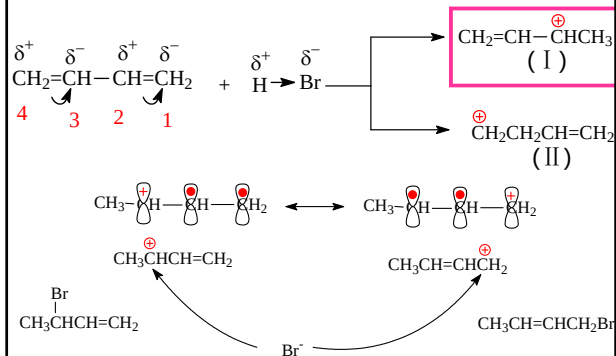
1. 共轭加成



共轭加成 —— 在加成反应中，共轭体系作为一个整体参与反应。

31

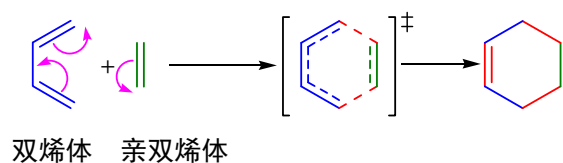
共轭二烯烃1, 4-加成的理论解释



32

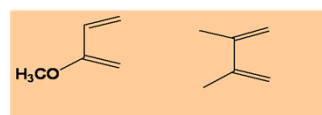
双烯合成——Diels-Alder反应

共轭二烯烃及其衍生物与含有碳碳双键、叁键等化合物进行1,4加成反应生成含六元环状的化合物。

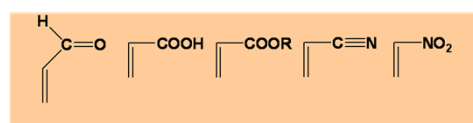


33

1) 具有供电子基团的双烯体有利于D-A反应

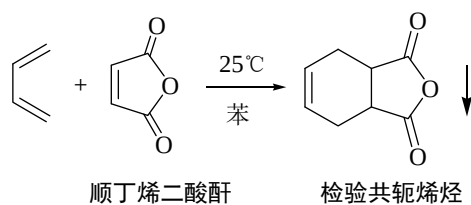


2) 具有吸电子基团的亲双烯体有利于D-A反应



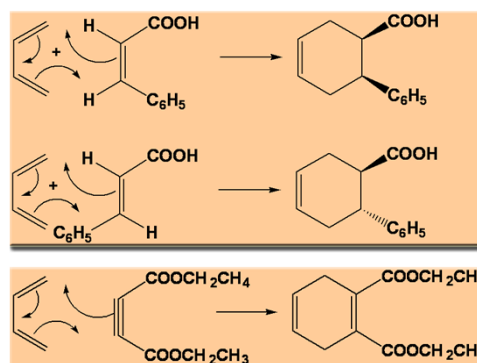
34

➤Diels-Alder反应的应用



35

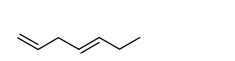
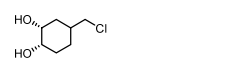
➤Diels-Alder反应是立体专一的顺式加成



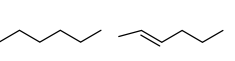
36

作业

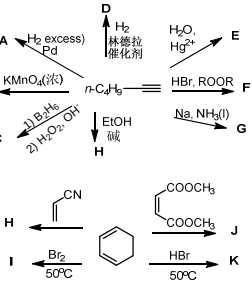
1, 由含3个碳及以下的原料合成下列物质



2, 鉴别下列四个化合物

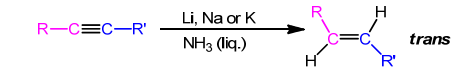


3, 完成下列反应

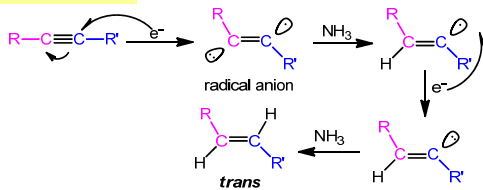


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本页仅作参考



Mechanism



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